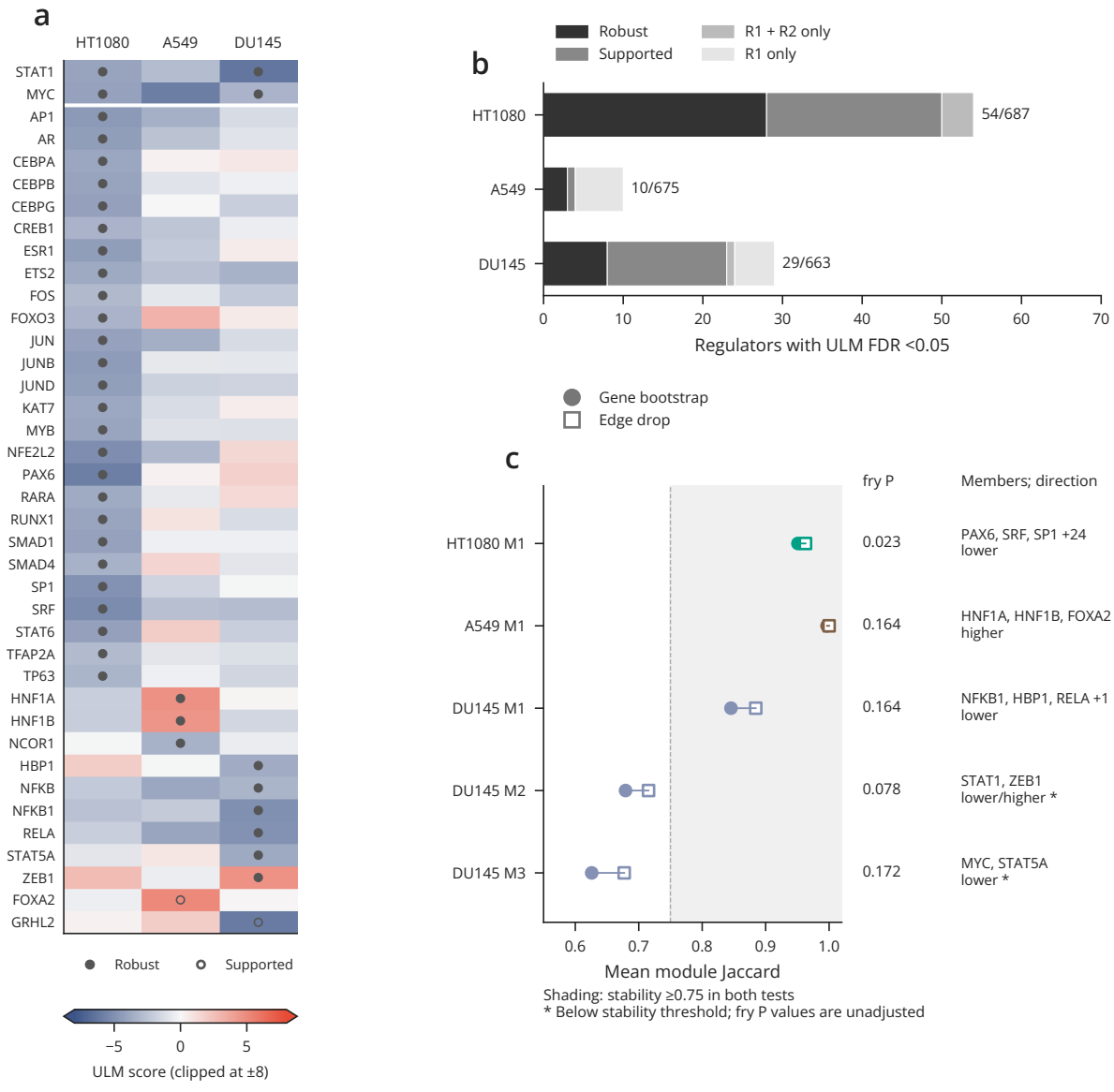
a
HT1080
A549
DU145
STAT1
MYC
AP1
AR
CEBPA
CEBPB
CEBPG
CREB1
ESR1
ETS2
FOS
FOXO3
JUN
JUNB
JUND
KAT7
MYB
NFE2L2
PAX6
RARA
RUNX1
SMAD1
SMAD4
SP1
SRF
STAT6
TFAP2A
TP63
HNF1A
HNF1B
NCOR1
HBP1
NFKB
NFKB1
RELA
STAT5A
ZEB1
FOXA2
GRHL2
Robust
Supported
−5
0
5
ULM score (clipped at ±8)
b
Robust
Supported
R1 + R2 only
R1 only
HT1080
A549
DU145
54/687
10/675
29/663
0
10
20
30
40
50
60
70
Regulators with ULM FDR <0.05
Gene bootstrap
Edge drop
c
fry P
Members; direction
HT1080 M1
0.023
PAX6, SRF, SP1 +24 lower
A549 M1
0.164
HNF1A, HNF1B, FOXA2 higher
DU145 M1
0.164
NFKB1, HBP1, RELA +1 lower
DU145 M2
0.078
STAT1, ZEB1 lower/higher *
DU145 M3
0.172
MYC, STAT5A lower *
0.6
0.7
0.8
0.9
1.0
Mean module Jaccard
Shading: stability ≥0.75 in both tests
* Below stability threshold; fry P values are unadjusted